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FUEL CELL COOLING SYSTEM

Abstract

A fuel cell cooling system includes a first cooling channel, a fuel cell, a second cooling channel, radiator, an intercooler, an intermediate heat exchanger including a first heat exchange channel and a second heat exchange channel, a parallel channel, and a valve. During power generation by the fuel cell, when the temperature of the coolant in the first cooling channel is lower than a first reference value, a first operation is performed in which coolant is circulated in the first cooling channel over a path passing through the fuel cell and the first heat exchange channel, and also coolant is circulated in the second cooling channel over a path passing through the radiator, the parallel channel and the intercooler. When the temperature of the coolant in the first cooling channel is higher than the first reference value, a second operation is performed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-032837 filed on Mar. 5, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The technology disclosed in the present specification relates to a fuel cell cooling system.

2. Description of Related Art

[0003] A fuel cell system disclosed in Japanese Unexamined Patent Application Publication No. 2002-33108 (JP 2002-33108 A) has a first cooling channel and a second cooling channel for circulating coolant. A fuel cell is provided on the first cooling channel. A radiator is provided on the second cooling channel. An intermediate heat exchanger is provided across the first cooling channel and the second cooling channel. The intermediate heat exchanger exchanges heat between the first cooling channel and the second cooling channel.

SUMMARY

[0004] In a fuel cell system, there are cases in which an intercooler is provided to cool compressed air that is supplied to a fuel cell. In the technology of JP 2002-33108 A, when the intercooler is provided on the second cooling channel, the coolant cooled by the radiator can be supplied to the intercooler and the intermediate heat exchanger. Heat generated in the fuel cell during power generation is transferred to the intermediate heat exchanger by the coolant flowing through the first cooling channel. The intermediate heat exchanger cools the coolant in the first cooling channel by heat exchange with the coolant in the second cooling channel (i.e., coolant cooled by the radiator). Also, the intercooler cools compressed air by heat exchange with the coolant in the second cooling channel (i.e., coolant cooled by the radiator).

[0005] When the fuel cell is started in a low-temperature environment, temperature of the fuel cell is low immediately after the start-up, and power generation efficiency is low. Further, when the fuel cell is started, coolant needs to be supplied to the intercooler. In the fuel cell system described above, when the coolant is supplied to the intercooler, the coolant is also supplied to the intermediate heat exchanger. This cools the coolant in the first cooling channel by the intermediate heat exchanger, and the temperature of the fuel cell does not readily rise. Accordingly, the power generation efficiency of the fuel cell does not readily rise. In this way, when the intercooler is provided on the second cooling channel, there is a problem in that the temperature of the fuel cell in a low-temperature environment does not readily rise. The present specification proposes technology capable of raising temperature of a fuel cell while supplying coolant to an intercooler.

First Aspect

[0006] A fuel cell cooling system disclosed in the present specification includes [0007] a first cooling channel through which coolant circulates, [0008] a fuel cell provided on the first cooling channel, [0009] a second cooling channel through which coolant circulates, [0010] a radiator provided on the second cooling channel, [0011] an intercooler provided on the second cooling channel, and [0012] an intermediate heat exchanger that includes a first heat exchange channel making up a portion of the first cooling channel and a second heat exchange channel making up a portion of the second cooling channel, and that performs heat exchange between the first heat exchange channel and the second heat exchange channel, [0013] in which [0014] the second cooling channel includes a parallel channel that is connected in parallel to the intermediate heat

exchanger and a valve that opens and closes the second heat exchange channel, [0015] during power generation by the fuel cell, when temperature of the coolant of the first cooling channel is lower than a first reference value, a first operation is performed in which coolant is circulated in the first cooling channel over a path passing through the fuel cell and the first heat exchange channel, and also coolant is circulated in the second cooling channel over a path passing through the radiator, the parallel channel, and the intercooler, in a state in which the second heat exchange channel is closed, and [0016] during power generation by the fuel cell, when the temperature of the coolant in the first cooling channel is higher than the first reference value, a second operation is performed in which the coolant is circulated in the first cooling channel over the path passing through the fuel cell and the first heat exchange channel, and also coolant is circulated in the second cooling channel over a path passing through the radiator, the second heat exchange channel, and the intercooler.

[0017] In this fuel cell cooling system, when the temperature of the coolant in the first cooling channel is lower than the first reference value during power generation by the fuel cell, the first operation is performed. In the first operation, the coolant is circulated in the second cooling channel over the path passing through the radiator, the parallel channel, and the intercooler, in the state in which the second heat exchange channel is closed. The intercooler can cool the compressed air by the coolant circulating in the second cooling channel. Also, the second heat exchange channel is closed, and accordingly the coolant does not flow into the intermediate heat exchanger. Thus, in the intermediate heat exchanger, heat exchange does not readily occur between the first heat exchange channel and the second heat exchange channel. As a result, the coolant in the first cooling channel is hardly cooled at all in the intermediate heat exchanger, and accordingly the temperature of the fuel cell can be efficiently raised. As described above, according to this fuel cell cooling system, the temperature of the fuel cell can be raised while the coolant is supplied to the intercooler.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0019] FIG. 1 is a block diagram of a fuel cell cooling system;

[0020] FIG. 2 is an explanatory diagram of a path;

[0021] FIG. 3 is an explanatory diagram of a path;

[0022] FIG. 4 is a flow chart of an operation selection process of the fuel cell cooling system; and

[0023] FIG. 5 is a flowchart of an operation selection process of the fuel cell cooling system.

DETAILED DESCRIPTION OF EMBODIMENTS

[0024] Following the above aspect 1, additional configurations of the fuel cell cooling system disclosed herein are described below.

Second Aspect

[0025] The fuel cell cooling system according to the first aspect, wherein the intercooler is provided in the parallel channel, and in the second operation, the coolant flows in parallel to the second heat exchange channel and the parallel channel.

Third Aspect

[0026] In the fuel cell cooling system according to the first or second aspect, in the first operation, when the temperature of the coolant in the first cooling channel is lower than the second reference value, the circulating speed of the coolant in the first cooling channel is slower than when the temperature of the coolant in the first cooling channel is higher than the second reference value.

Fourth Aspect

[0027] In the fuel cell cooling system according to any one of the first to third aspects, in the second operation, when the temperature of the coolant in the first cooling channel is higher than the third reference value, the flow rate of the coolant in the second heat exchange channel is made faster than when the temperature of the coolant in the first cooling channel is lower than the third reference value.

[0028] According to the second aspect, since the coolant cooled by the radiator flows in parallel to the intercooler and the second heat exchange channel, the intercooler and the intermediate heat exchanger can be operated efficiently.

[0029] According to the third aspect, when the temperature of the coolant in the first cooling channel is low in the first operation, the temperature of the fuel cell can be raised faster.

[0030] According to the fourth aspect, when the temperature of the coolant in the first cooling channel is high in the second operation, the fuel cell can be cooled more effectively.

[0031] The fuel cell cooling system **100** shown in FIG. **1** is mounted on a device (for example, a fuel cell electric vehicle) using a fuel cell as a power source. The fuel cell cooling system **100** includes a first cooling channel **12**, a second cooling channel **22**, and an intermediate heat exchanger **30**.

[0032] The intermediate heat exchanger **30** has a first heat exchange channel **30a** and a second heat exchange channel **30b**. The first heat exchange channel **30a** is a part of the first cooling channel **12**, and the second heat exchange channel **30b** is a part of the second cooling channel **22**. The intermediate heat exchanger **30** causes heat exchange between the first heat exchange channel **30a** and the second heat exchange channel **30b**. That is, the intermediate heat exchanger **30** causes heat exchange between the first cooling channel **12** and the second cooling channel **22**.

[0033] A fuel cell **14** is provided in the first cooling channel **12**. The fuel cell **14** is supplied with compressed air from an air compressor (not shown) and hydrogen from a tank (not shown). The fuel cell **14** generates electric power by reacting oxygen with hydrogen, and supplies electric power to a motor (not shown). The fuel cell **14** is cooled by the coolant in the first cooling channel **12**.

[0034] The first cooling channel **12** is an annular channel. The first cooling channel **12** is provided with a pump **16** and temperature sensors **18a**, **18b**. The pump **16** delivers the coolant from the installation position toward the intermediate heat exchanger **30**. When the pump **16** is operated, the coolant circulates in the first cooling channel **12**.

[0035] The temperature sensor **18a** is provided in the first cooling channel **12** upstream of the fuel cell **14**. The temperature sensor **18a** detects the temperature of the coolant flowing into the fuel cell **14** (hereinafter referred to as FC inlet temperature). The temperature sensor **18b** is provided downstream of the fuel cell **14** in the first cooling channel **12**. The temperature sensor **18b** detects the temperature of the coolant discharged from the fuel cell **14** (hereinafter referred to as FC outlet temperature).

[0036] A radiator **24** is provided in the second cooling channel **22**. The radiator **24** cools the coolant flowing in the second cooling channel **22** by heat exchange with the outside air.

[0037] The second cooling channel **22** includes a channel **22a**, a channel **22b**, a parallel channel **34**, and a three-way valve **38**. The upstream end of the channel **22a** is connected to the downstream end of the coolant channel in the radiator **24**. The downstream end of the channel **22a** is connected to the upstream end of the second heat exchange channel **30b** and the upstream end of the parallel channel **34** via the three-way valve **38**. The parallel channel **34** is arranged in parallel to the second heat exchange channel **30b**. An upstream end of the channel **22b** is connected to a downstream end of the second heat exchange channel **30b**. The downstream end of the channel **22b** is connected to the upstream end of the coolant channel in the radiator **24**.

[0038] The three-way valve **38** opens and closes the second heat exchange channel **30b**. According to the opening degree of the three-way valve **38**, the flow rate of the coolant flowing in each of the second heat exchange channel **30b** and the parallel channel **34** is adjusted.

[0039] An intercooler **32** is provided in the parallel channel **34**. The intercooler **32** cools the compressed air supplied to the fuel cell **14** by heat exchange with the coolant flowing through the parallel channel **34**.

[0040] The fuel cell cooling system **100** includes a pump **26**, a temperature sensor **28a**, and a temperature sensor **28b**. The pump **26** is provided in the channel **22a**. The pump **26** delivers the coolant from the installation position toward the intermediate heat exchanger and the intercooler **32**. When the pump **26** is operated, the coolant circulates in the second cooling channel **22**.

[0041] The temperature sensor **28a** is provided in the channel **22a**. The temperature sensor **28a** detects a temperature of the coolant discharged from the radiator **24** (hereinafter, referred to as a radiator outlet temperature). The temperature sensor **28b** is provided in the channel **22b**. The temperature sensor **28b** detects a temperature of the coolant flowing into the radiator **24** (hereinafter, referred to as a radiator inlet temperature).

[0042] The fuel cell cooling system **100** includes a control device **40**. The control device **40** controls the pump **16**, the pump **26**, and the three-way valve **38** based on the temperature of the coolant in the first cooling channel **12** and the temperature of the coolant in the second cooling channel **22**.

[0043] The fuel cell **14** is supplied with compressed air from an air compressor (not shown) and hydrogen from a tank (not shown). The fuel cell **14** generates electric power by reacting oxygen with hydrogen. During power generation of the fuel cell **14**, the fuel cell cooling system **100** may perform the warm-up control and the normal control described below.

[0044] The warm-up control is executed when the temperature of the fuel cell **14** is low. In the warm-up control, the control device **40** activates the pump **16**. When the pump **16** is operated, the coolant circulates in the first cooling channel **12** in the order of the intermediate heat exchanger **30** (that is, the first heat exchange channel **30a**) and the fuel cell **14**. Therefore, the heat generated by the power generation of the fuel cell **14** is transferred to the first heat exchange channel **30a** by the coolant in the first cooling channel **12**.

[0045] In the warm-up control, the control device **40** operates the pump **26**. Furthermore, the control device **40** controls the three-way valve **38** with the second heat exchange channel **30b** closed. Therefore, the coolant circulates in the path **101** indicated by the arrow in FIG. 2. In the path **101**, the coolant delivered by the pump **26** circulates through the intercooler **32** (i.e., the parallel channel **34**), the channel **22b**, the radiator **24**, and the channel **22a** through the second cooling channel **22**. In this case, the coolant does not flow in the second heat exchange channel **30b**. Therefore, in the intermediate heat exchanger **30**, the coolant in the first heat exchange channel **30a** is hardly cooled. In the intercooler **32**, the compressed air supplied to the fuel cell **14** is cooled by heat exchange with the coolant flowing through the parallel channel **34**.

[0046] As described above, in the warm-up control, the compressed air supplied to the fuel cell **14** is cooled by the intercooler **32**, while the coolant in the first cooling channel **12** is hardly cooled in the intermediate heat exchanger **30**. Therefore, the temperature of the fuel cell **14** tends to increase.

[0047] Normal control is performed when the temperature of the fuel cell **14** is high. In normal control, the control device **40** activates the pump **16**. Therefore, similarly to the warm-up control, the coolant circulates in the first cooling channel **12**. Therefore, the heat generated by the power generation of the fuel cell **14** is transferred to the intermediate heat exchanger **30** (that is, the first heat exchange channel **30a**) by the coolant in the first cooling channel **12**.

[0048] In normal control, the control device **40** activates the pump **26**. In addition, the control device **40** adjusts the opening degree of the three-way valve **38**, and controls the flow of the coolant in parallel with the second heat exchange channel **30b** and the parallel channel **34**. Therefore, the coolant circulates in the path **102** indicated by the arrow in FIG. 3. In path **102**, the coolant pumped by pump **26** flows in parallel to intercooler **32** (i.e., parallel channel **34**) and intermediate heat exchanger **30** (i.e., second heat exchange channel **30b**). The coolant that has passed through the intermediate heat exchanger **30** and the intercooler **32** then flows to the radiator **24**. The radiator **24**

cools the coolant in the second cooling channel 22 by heat exchange with the outside air. Therefore, the coolant cooled by the radiator 24 flows to the intermediate heat exchanger 30 and the intercooler 32. The intermediate heat exchanger 30 cools the coolant in the first cooling channel 12 by heat exchange between the first cooling channel 12 and the second cooling channel 22. Since the coolant in the second cooling channel 22 is cooled by the radiator 24, the intermediate heat exchanger 30 can efficiently cool the coolant in the first cooling channel 12. The intercooler 32 cools the compressed air supplied to the fuel cell 14 by heat exchange with the coolant flowing through the parallel channel 34.

[0049] As described above, in the normal control, the coolant in the first cooling channel 12 can be effectively cooled in the intermediate heat exchanger 30 while the compressed air supplied to the fuel cell 14 is cooled by the intercooler 32.

[0050] FIG. 4 illustrates in more detail the processing performed by the control device 40 during operation of the fuel cell 14.

[0051] In S2, the control device 40 measures the outside air temperature by a temperature sensor (not shown). The control device 40 determines whether the outside air temperature is equal to or lower than the cold determination temperature T1. When the vehicle is stopped, the temperature of the coolant substantially coincides with the outside air temperature. Therefore, S2 process is equivalent to determining whether the temperature of the coolant flowing through the first cooling channel 12 is equal to or lower than the cold determination temperature T1.

[0052] When the outside air temperature is equal to or lower than the cold determination temperature T1 (YES in S2), the control device 40 performs the cold/warm-up control in S4. In the cold/warm-up control, the control device 40 performs the above-described warm-up control. In the cold/warm-up control, the control device 40 controls the discharge flow rate of the pump 26 in accordance with the radiator outlet temperature and the heat capacity of the compressed air flowing into the intercooler 32. Further, in the cold/warm-up control, the control device 40 controls the output current of the fuel cell 14 and the discharge flow rate of the pump 16 in accordance with FC inlet temperature. At this time, the discharge flow rate of the pump 16 is controlled to a low value. As described above, in the warm-up control, the temperature of the fuel cell 14 tends to increase. In particular, in the cold/warm-up control, since the discharge flow rate of the pump 16 is low, it is difficult to transfer heat from the fuel cell 14 to the coolant. Therefore, the temperature of the fuel cell 14 tends to increase. As described above, when the air temperature is lower than or equal to the cold determination temperature T1, the temperature of the fuel cell 14 is rapidly increased by performing the cold/warm-up control. This increases the power generation efficiency of the fuel cell 14.

[0053] When FC outlet temperature is lower than the cold/warm-up control end temperature T2, the control device 40 performs cold/warm-up control (that is, NO in YES, S8 in S6) until the power of the vehicles is turned OFF. The control device 40 performs S12 when FC outlet temperature exceeds the cold/warm-up control end temperature T2 (NO in S6).

[0054] If the outside air temperature is higher than the cold determination temperature T1 in S2 (NO in S2), the control device 40 determines whether FC outlet temperature is lower than the warm-up control end temperature T3 in S10. That is, the control device 40 determines whether the temperature of the coolant flowing through the first cooling channel 12 is lower than the warm-up control end temperature T3. The warm-up control end temperature T3 is a temperature higher than the cold determination temperature T1 and the cold/warm-up control end temperature T2.

[0055] When FC outlet temperature is lower than the warm-up control end temperature T3 (YES in S10), the control device 40 performs the normal warm-up control in S12. In the normal warm-up control, the control device 40 performs the above-described warm-up control. In the normal warm-up control, the control device 40 controls the discharge flow rate of the pump 26 in accordance with the radiator outlet temperature and the heat capacity of the compressed air flowing into the intercooler 32. In the normal warm-up control, the control device 40 controls the output current of

the fuel cell **14** and the discharge flow rate of the pump **16** in accordance with FC inlet temperature. In the normal warm-up control, the discharge flow rate of the pump **16** is controlled to a high value (that is, a value higher than **S4**). As described above, in the warm-up control, the temperature of the fuel cell **14** tends to increase. However, in the normal warm-up control, since the discharge flow rate of the pump **16** is high, the temperature of the fuel cell **14** is less likely to increase than in the cold-warm-up control. In this way, in a state where the temperature of the fuel cell **14** is high to some extent, the power generation amount of the fuel cell **14** is lowered and the flow rate of the coolant is increased more than in the cold/warm-up control, thereby reducing the temperature gradient in the fuel cell **14**.

[0056] When FC outlet temperature is lower than the warm-up control end temperature **T3**, the control device **40** performs the normal warm-up control until the power of the vehicle is turned OFF (that is, YES, **S14** and NO in **S10**). The control device **40** performs **S16** when FC outlet temperature exceeds the warm-up control end temperature **T3** (NO in **S10**).

[0057] In **S16**, the control device **40** performs the above-described normal control. At this time, the opening degree of the three-way valve **38** is fixed. In this way, when FC outlet temperature is high (that is, when the temperature of the fuel cell **14** is high), the fuel cell **14** is effectively cooled by the normal control. This suppresses excessive temperature rise of the fuel cell **14**.

[0058] When FC outlet temperature is higher than the warm-up control restart temperature **T4**, the control device **40** performs the normal control (that is, NO in NO, **S20** in **S18**) until the power of the vehicle is turned OFF. The warm-up control restart temperature **T4** is a temperature lower than the warm-up control end temperature **T3**. When FC outlet temperature becomes lower than the warm-up control restart temperature **T4** (YES in **S18**), the control device **40** performs the normal warm-up control in **S12**.

[0059] In the above-described embodiment, the opening degree of the three-way valve **38** was fixed by **S16**. However, the opening degree of the three-way valve **38** may be adjusted based on the calorific value of the fuel cell **14**, FC outlet temperature, and the heat capacity of the compressed air flowing into the intercooler **32**. In this instance, the control device **40** executes the process of FIG. 5 (i.e., from **S22** to **S30**) during execution of **S16** (i.e., during execution of the normal control).

[0060] In **S22**, the control device **40** determines the power of the radiator **24** according to the amount of heat generated by the fuel cell **14** and the vehicle speed. Therefore, the radiator **24** appropriately cools the coolant.

[0061] Next, the control device **40** determines a quantity **Q1** (hereinafter, referred to as a flow rate **Q1**) of the coolant flowing to the intercooler **32** in accordance with the heat capacity of the compressed air flowing into the intercooler **32** in **S24** and the radiator outlet temperature. In addition, the control device **40** determines a quantity **Q2** (hereinafter, referred to as a flow rate **Q2**) of the coolant flowing to the intermediate heat exchanger **30** in accordance with the quantity of heat generated by the fuel cell **14**.

[0062] Next, the control device **40** determines whether FC outlet temperature is lower than the target temperature **T5** in **S26**. That is, the control device **40** determines whether the temperature of the coolant flowing through the first cooling channel **12** is lower than the target temperature **T5**. The control device **40** performs **S28** when FC outlet temperature is equal to or lower than the target temperature **T5** (when **S26** is YES).

[0063] In **S28**, the control device **40** adjusts the opening degree of the three-way valve **38** and the output of the pump **26** based on the flow rates **Q1**, **Q2** (i.e., the flow rate) determined by **S24**. Since the flow rate **Q2** is determined according to the calorific value of the fuel cell **14**, heat is efficiently exchanged between the coolant flowing through the intermediate heat exchanger **30** and the coolant flowing through the first cooling channel **12**. Therefore, the fuel cell **14** is efficiently cooled. Further, the flow rate **Q1** was determined according to the heat capacity of the compressed air flowing into the intercooler **32** and the radiator outlet temperature. As a result, in the intercooler **32**,

the compressed air supplied to the fuel cell **14** is cooled more effectively.

[0064] The control device **40** performs **S30** when FC outlet temperature exceeds the target temperature **T5** (when **S26** is NO).

[0065] In **S30**, the control device **40** determines the flow rate **Q3** (i.e., the flow speed) of the coolant flowing to the intermediate heat exchanger **30** so as to satisfy the following relational expression.

[00001] $Q3 = Q2 \cdot \text{Math.} \{1 + a \cdot \text{Math.} (\text{FCoutlettemperature} - T5)\}$ [0066] The value of the flow rate **Q3** is a value larger than the flow rate **Q2**.

[0067] **S30** then adjusts the opening of the three-way valve **38** and the power of the pump **26** based on the determined flow rate **Q1** and flow rate **Q3**. Since the flow rate **Q3** is larger than the flow rate **Q2**, in **S30**, heat is more likely to be exchanged between the coolant flowing through the intermediate heat exchanger **30** and the coolant flowing through the first cooling channel **12** than **S28**. Therefore, the fuel cell **14** can be cooled more effectively.

[0068] In the above-described embodiment, the three-way valve **38** is provided at a connecting portion between the channel **22a**, the parallel channel **34**, and the upstream end of the second heat exchange channel **30b**. However, the three-way valve **38** may be provided at a connecting portion between the channel **22b**, the parallel channel **34**, and the downstream end of the second heat exchange channel **30b**.

[0069] In the above-described embodiment, the channel changing device for changing the path of the coolant in the second cooling channel **22** by the three-way valve **38** is configured, the channel changing device may be configured by other devices. Further, in the embodiment, the three-way valve **38** is a solenoid valve, but the channel changing device may be configured by a device that does not use electricity. For example, a channel changing device may be configured by a device that switches a channel by thermal expansion of a material, such as a thermostat.

[0070] In the embodiment, the intercooler **32** is provided in the parallel channel **34**. However, the intercooler **32** may be provided in the channel **22a**. Further, the intercooler **32** may be a channel **22b**, and may be provided between the parallel channel **34** and the radiator **24**. That is, the coolant may flow in series with the intermediate heat exchanger **30** and the intercooler **32**.

[0071] In **S4**, **S12** of the above-described embodiment, the control device **40** controls the discharge flow rate of the pump **16** in accordance with FC inlet temperature. However, in **S4**, **S12**, the discharge flow rate of the pump **16** may be mapped.

[0072] In **S4**, **S12** of the above-described embodiment, the discharge flow rate of the pump **26** is controlled in accordance with the radiator outlet temperature and the heat capacity of the compressed air flowing into the intercooler **32**. However, in **S4**, **S12**, the discharge flow rate of the pump **26** may be mapped.

[0073] In the above-described embodiment, the output of the fan of the radiator **24** is determined in accordance with the amount of heat generated by the fuel cell **14** and the vehicle speed. However, the amount of heat dissipation of the radiator **24** may be calculated from the radiator outlet temperature and the radiator inlet temperature, and the output of the fan of the radiator **24** may be determined from the comparison between the amount of heat generation of the fuel cell **14** and the amount of heat dissipation of the radiator **24**.

[0074] The cold/warm-up control and the normal warm-up control of the embodiment are examples of the first operation. Further, the normal control is an example of the second operation.

[0075] The warm-up control end temperature **T3** of the embodiment is an example of the first reference value. The cold determination temperature **T1** of the embodiment is an example of the second reference value. The target temperature **T5** is an exemplary third reference value.

[0076] While the embodiments have been described in detail above, these are merely illustrative and do not limit the scope of the claims. The technology described in the claims includes various modifications and alterations of the specific examples described above. The technical elements described in this specification or in the drawings may be used alone or in various combinations,

and are not limited to the combinations described in the claims at the time of filing. Further, the technology illustrated in the present specification or the drawings achieves a plurality of objects at the same time, and has technical usefulness by achieving one of the objects.

Claims

1. A fuel cell cooling system comprising: a first cooling channel through which coolant circulates; a fuel cell provided on the first cooling channel; a second cooling channel through which coolant circulates; a radiator provided on the second cooling channel; an intercooler provided on the second cooling channel; and an intermediate heat exchanger that includes a first heat exchange channel making up a portion of the first cooling channel and a second heat exchange channel making up a portion of the second cooling channel, and that performs heat exchange between the first heat exchange channel and the second heat exchange channel, wherein the second cooling channel includes a parallel channel that is connected in parallel to the intermediate heat exchanger and a valve that opens and closes the second heat exchange channel, during power generation by the fuel cell, when temperature of the coolant of the first cooling channel is lower than a first reference value, a first operation is performed in which coolant is circulated in the first cooling channel over a path passing through the fuel cell and the first heat exchange channel, and also coolant is circulated in the second cooling channel over a path passing through the radiator, the parallel channel, and the intercooler, in a state in which the second heat exchange channel is closed, and during power generation by the fuel cell, when the temperature of the coolant in the first cooling channel is higher than the first reference value, a second operation is performed in which the coolant is circulated in the first cooling channel over the path passing through the fuel cell and the first heat exchange channel, and also coolant is circulated in the second cooling channel over a path passing through the radiator, the second heat exchange channel, and the intercooler.
 2. The fuel cell cooling system according to claim 1, wherein: the intercooler is provided on the parallel channel; and in the second operation, the coolant flows in parallel over the second heat exchange channel and the parallel channel.
 3. The fuel cell cooling system according to claim 1, wherein, in the first operation, when the temperature of the coolant in the first cooling channel is lower than a second reference value, a circulating rate of the coolant in the first cooling channel is reduced as compared to when the temperature of the coolant in the first cooling channel is higher than the second reference value.
 4. The fuel cell cooling system according to claim 1, wherein, in the second operation, when the temperature of the coolant in the first cooling channel is higher than a third reference value, a flow rate of the coolant in the second heat exchange channel is increased as compared to when the temperature of the coolant in the first cooling channel is lower than the third reference value.
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